DIPLOMA PROGRAMME PRACTICE · SL · SET B

Chemistry / Paper 2

Worked mark scheme · 50 marks · 90 minutes

Award one mark per marking point. Credit equivalent reasoning and suitable units. Multiple-choice decisions carry one mark each.

PAPER 2 · QUESTION 1 [10]

Layered carbon in an electrolytic cell

(a) Compare the two distances numerically and explain what the difference indicates about the interactions. [3]

1 335/142≈2.36; the interlayer spacing is about 2.36 times the bond length.

1 Within a layer, strong covalent C–C bonds form a network.

1 Between layers the attractions are weaker London forces; the dotted lines do not represent further covalent C–C bonds.

(b) Explain how graphite can conduct parallel to its layers and why the layers can slide. [2]

1 Each carbon bonds to three neighbours, leaving electrons delocalized across its layer.

1 Delocalized electrons carry current, while weak interlayer attractions allow layers to slip.

(c) Write both electrode equations for molten $PbBr_2$ and identify the electrode where oxidation occurs. [3]

1 Cathode: $Pb^{2+}+2e^{-}\rightarrow Pb$.

1 Anode: $2Br^{-}\rightarrow Br_2+2e^{-}$.

1 Oxidation is loss of electrons by bromide at the anode.

(d) Determine the mass of lead when 0.0250 mol Br_2 is collected under conditions of complete product recovery. Use $M(Pb)=207.2\ g\ mol^{-1}$. [2]

1 The electrode reactions require equal amounts of Pb and Br_2, hence 0.0250 mol Pb.

1 $m=0.0250\times207.2=5.18$ g.

PAPER 2 · QUESTION 2 [10]

Analysing a gas blend

(a) Write balanced equations for complete combustion of each hydrocarbon. [2]

1 $CH_4+2O_2\rightarrow CO_2+2H_2O$.

1 $2C_2H_6+7O_2\rightarrow 4CO_2+6H_2O$.

(b) Determine the volume percentage of each hydrocarbon. [3]

1 Let x be ethane volume, so methane volume is 100.0−x.

1 CO_2 volume=(100.0−x)+2x=145.0, hence x=45.0 cm³.

1 45.0% ethane and 55.0% methane by volume.

(c) Calculate the oxygen volume consumed and the volume remaining if the original oxygen supply was 300.0 cm³. [3]

1 Methane uses 2×55.0=110.0 cm³ O_2.

1 Ethane uses 3.5×45.0=157.5 cm³ O_2; total=267.5 cm³.

1 Remaining oxygen=32.5 cm³.

(d) A leak removes some CO_2 before its measurement. Explain the error in the inferred ethane percentage and one way to check the collection apparatus. [2]

1 A lower CO_2 yield is interpreted as less of the two-carbon gas; ethane fraction is underestimated.

1 Pressure-test the sealed apparatus or pass a known gas volume through and measure its recovery.

PAPER 2 · QUESTION 3 [10]

Ammonia and a strong acid

(a) Write the proton-transfer equation using H_3O^+ and identify the two conjugate acid–base pairs. [3]

1 $NH_3+H_3O^+\rightarrow NH_4^++H_2O$.

1 NH_4^+/NH_3 is one conjugate pair.

1 H_3O^+/H_2O is the other.

(b) Determine the limiting reactant and the amount of excess strong acid. [3]

1 $NH_3=0.0400\times0.150=0.00600$ mol and $HCl=0.0300\times0.250=0.00750$ mol.

1 The 1:1 transfer makes NH_3 limiting.

1 Excess strong acid=0.00150 mol.

(c) Determine the final pH. Explain why simply averaging the initial pH values would fail. [3]

1 $[H^+]=0.00150/0.0700=0.02143$ mol dm^{-3}.

1 pH=1.67.

1 Acid–base reaction changes amounts; pH is logarithmic and cannot be averaged by volume.

(d) Explain the origin of the new N–H bonding pair when NH_3 accepts a proton. [1]

1 Both electrons originate from the nitrogen lone pair, forming a coordinate covalent bond.

PAPER 2 · QUESTION 4 [10]

Reversing a dissolution measurement

(a) Determine the heat change of the solution and the molar enthalpy of dissolution. [3]

1 $q_{solution}=100.0\times4.18\times(-3.30)=-1.379$ kJ.

1 $n(salt)=5.00/80.0=0.0625$ mol.

1 $\Delta H_{diss}=+1.379/0.0625=+22.1$ kJ mol^{-1}.

(b) Explain the sign using the competing energy changes when an ionic solid dissolves. [2]

1 Separating ions from their lattice requires energy.

1 Hydration of ions releases energy, but the net endothermic result shows the energy requirement exceeds the energy released.

(c) Each temperature reading has maximum uncertainty ±0.05 °C. With other quantities exact, estimate the maximum percentage and absolute uncertainties in ΔH. [3]

1 Maximum uncertainty in the fall is ±0.10 °C.

1 Relative uncertainty=0.10/3.30×100=3.03%.

1 Absolute enthalpy uncertainty≈0.67 kJ mol^{-1}; accept ±0.7.

(d) If heat enters from the warmer surroundings during dissolution, explain how the reported ΔH changes. [2]

1 Heat entering offsets some of the temperature fall.

1 The inferred positive dissolution enthalpy is smaller than its true magnitude.

PAPER 2 · QUESTION 5 [10]

Following a four-carbon alcohol

(a) Identify the alcohol and the ketone by name and condensed structural formula. [3]

1 Butan-2-ol, $CH_3CH(OH)CH_2CH_3$.

1 Butan-2-one, $CH_3COCH_2CH_3$.

1 The OH-bearing carbon is bonded to two carbon groups, so this secondary alcohol gives a ketone.

(b) Write the oxidation equation using [O] and show the change in oxidation state of the carbon bearing OH. [3]

1 $CH_3CH(OH)CH_2CH_3+[O]\rightarrow CH_3COCH_2CH_3+H_2O$.

1 In the alcohol that carbon has one C–O and one C–H bond; contributions +1 and −1 give 0.

1 In the ketone the C=O contributes +2, a rise from 0 to +2.

(c) Give the names and condensed formulas of the two successive oxidation products of its straight-chain primary alcohol isomer. [2]

1 Butanal, $CH_3CH_2CH_2CHO$, from butan-1-ol.

1 Butanoic acid, $CH_3CH_2CH_2COOH$, on further oxidation.

(d) Explain why butan-2-ol has a higher normal boiling temperature than an ether of similar molar mass that lacks an O–H bond. [2]

1 Butan-2-ol molecules can donate and accept intermolecular hydrogen bonds.

1 A pure ether lacks an O–H donor; its intermolecular attractions are generally weaker here, requiring less energy for vaporization.